

Experiment 02: AI Problem Formulation and Solving

Learning Objective: Student should be able to apply AI problem formulation approach to solve any problem.

Tools: Python under Windows or Linux environment/Online Platform

Theory:

Formulating problems : A problem is defined by four items:

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1. initial state e.g., "at Arad"
2. actions or successor function or operators

$S(x)$ = set of action–state pairs

– e.g., $S(Arad) = \{Arad \rightarrow Zerind, Zerind \rightarrow \dots\}$

3. goal test, can be

- explicit, e.g., $x = \text{"at Bucharest"}$
- implicit, e.g., $Checkmate(x)$

4. path cost function (additive)

- e.g., sum of distances, number of actions executed, etc.
- $c(x,a,y)$ is the step cost, assumed to be ≥ 0

A solution is a sequence of actions leading from the initial state to a goal state

Problem Formulation:

(1) Initial state:

(2) Actions or successor function or operators :

(3) Goal test

(4) Path cost function

(5) Solution (Please include state space search graph)

Implementation:

Result and discussion:

Learning Outcomes: Students should have the ability to

LO1: Formulate the problem using AI Approach.

LO2 : Solve the problem using AI Approach.

Course Outcomes: Upon completion of the course students will be able to evaluate various problem solving methods for an agent to find a sequence of actions to reach the goal state.

Conclusion:

Viva Questions:

Q. 1 Define problem solving agent with example.

Q.2 Illustrate 3 problems can be solved using problem formulation approach.

Q.3 Discuss various items or elements to be considered during problem formulation.

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				